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Paper 4: Selection Functionals & Paper 5: Memory Rank - Response Addendum

Benjamin D. Summers <bsummers@adagiollc.com> Fri, Sep 18, 2026 at 1:09 AM
To: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>, Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Hi Ezequiel and Sara,

I have a follow-up to my Paper 4 Selection Functionals and Paper 5 Memory Rank responses:

Selection functionals. Revision 2.3 of the manuscript is on SSRN (https://dx.doi.org/10.2139/ssrn.6922758). No definition, result statement, proof, table or displayed equation changed since my initial response, and every section and result number in the initial response still points to the same item. I reordered or restated 72 sentences of the body and the third sentence of the abstract so that each states its content in plain order. The paper now cites only the three papers that precede it in the series. The citations of the projection paper and the balance-law paper are removed, and the seven sentences that carried them (in the introduction, Remark 2.2, Sections 7.1, 7.3 and 8.3, Remark 7.6 and the bridge paragraph of Section 11) are restated or dropped with no change to any claim. The sentence in my initial response about updating the projection-paper citation no longer applies. My initial response said the Finding 2 identity and the Finding 3 construction are in the paper with attribution.

Memory rank. Revision 2.1 of the manuscript is on SSRN (https://dx.doi.org/10.2139/ssrn.7119779). No definition, proof, numbered result or displayed equation changed in content since my initial response, and every section and result number in the initial response still points to the same item; Proposition 5.3 now describes the companion deficit norms as norms of Δ_V integrated over time rather than weighted L² norms, and three clauses are made precise (the block masses per time in Example 4.10, the ε² rate of the distance to Flat in Example 8.5, e_b defined in Remark 3.6). The paper now cites only the four papers that precede it in the series. The citation of the balance-law paper is removed, and the three sentences that carried it (the program list in Section 1, the last sentence of Example 5.2, and Open Problem (v)) are restated; the clause in Example 5.2 and Open Problem (v) that the metric charge does not decompose algebraically, which no result of the paper proves, is withdrawn with them. The conditioning bound of Remark 3.6 and Example 8.5 is written K, since κ is the compression, so my initial response's F_κ is the paper's F_K. The citation of Proposition 3.6 of the Koopman paper now carries that proposition's scope, strongly measurable Bochner-integrable generators. Five sentences of the body are restated in plain order, and the significance statement is removed. My initial response said your D_max argument, joint-law obstruction, witness, split-and-lump construction, identification of the re-dressing group and the question behind Proposition 4.11 are in the paper with attribution.

All six papers credit you both for critical review of the manuscripts, and none carries an itemized attribution.

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On Tue, Sep 15, 2026 at 10:08 PM Benjamin D. Summers <bsummers@adagiollc.com> wrote:

> Hi Ezequiel and Sara,
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> Thank you both for your reviews (both attached). The revised manuscript has been uploaded to SSRN (https://dx.doi.org/10.2139/ssrn.6922758). Again, you both reviewed the same text without conferring, and again, you both converged on the same points: the inference that opened Section 7.2, the standing of Theorems A–C relative to standard importance-sampling theory, the calibration section, and the reading of Proposition 2.3. Below are my responses in manuscript order, starting with the one that explains most of the others. As previously, where you both raised a point, I answer it once. Section and result numbers are those of the revision; where they differ from the July version I give both.
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> Ezequiel, thank you for the verification record. I recomputed every figure in your report. The closed forms agree to the printed digit: the ratio-weight table and the 9.8×10^{49}, the factor 2.2, the exponent 4.2×10^{-16}, the M = 8 eigenvalues and conditional-mean coefficients (your triple (2,4,6) is (3,5,7) in the paper's 1-based indexing), the

Bhattacharyya table, and all twenty entries of the temperature-schedule table. The sweep of Finding 5 agrees to the resolution of your grid: your k = 2.75 is the 0.05-grid minimizer (the continuous minimizer is 2.762), $\sigma^2(2.75) = 20.43$, the bottom at $v^\dagger$ is flat ($\sigma^2(12.36) = 4.9988$ against $\sigma^2(12.39) = 4.9987$), and 4.08 is the ratio at your points. I could not reproduce two Monte Carlo figures. Your pooled 700-replication value 3.44 at $N = 10^4$: 10,000 replications give me 4.02 with bootstrap standard error 0.05, every 700-replication chunk lies in [3.79, 4.24], and the bootstrap puts a value at or below 3.44 under 3×10^{-4}. Your single batch 7.53 lies above the maximum of my 200 batches of 50, which is 6.88; the bootstrap puts it at one batch in 2,500. Your ten-batch range [3.30, 4.57] sits between the 11th and 84th percentiles of those 200 batches, and your plug-in draw 833.6 sits inside twenty draws spanning 435 to 1684. Both conclusions you draw from them hold at my numbers: 4.02 is 28% below the asymptote 5.56, and 95% of plug-in draws fall below the exact 1372.4. Table 1 and the plug-in 623 of the July version reproduce to the digit from my archived script and seed 20260717; the manuscript did not say so, and the revision deposits the code and seeds. Your Finding 2 identity and your Finding 3 construction are in the paper with attribution.

The identity in Ezequiel's Finding 2 explains most of the other changes. For constant drift divergence, the selection weight is the surrogate's own Euler path density raised to the power λ, up to a constant, so the selected measure is a product of two path densities, $Q \propto \Pi \cdot p_\nu^\lambda$. For a Gaussian proposal and surrogate the precisions add. The interior rows of $K_\Pi + \lambda K_\nu$ are those of an Ornstein–Uhlenbeck Euler chain with $k_{eff}^2/\sigma_{eff}^2 = k_\Pi^2/\sigma_\Pi^2 + \lambda k_\nu^2/\sigma_\nu^2$ exactly and $1/\sigma_{eff}^2 = 1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2 + O(\delta t)$; only the last row differs. This is Proposition 3.5 and Corollary 3.6. In the worked example $\sigma_{eff}^2 = 0.4992$ and $k_{eff} = 1.58$, and the increment variance under Q is 0.0101 against 0.0201 under Π. I had not seen this, and it changes what the paper says in four places. The chi-square blow-up under refinement is quadratic-variation mismatch between two Gaussian chains, not the pointwise degeneracy of Proposition 2.3; Section 9.4 now says so and gives the Girsanov ratio weight on the same grids ($\chi^2 = 0.1645$ to 0.1562) as the control. The temperature is a diffusion-coefficient control, and $\sigma_{eff} < \sigma_\Pi$ for every $\lambda > 0$ (Kolmogorov–Szegő), so the selected measure is singular to the proposal in the continuum at every fixed temperature. The transfer hypothesis (D1) can hold under refinement only if the selected quadratic variation matches the target's, which is the design rule of Section 7.5. The construction differs from Girsanov reweighting in that it multiplies densities where that literature divides them; that is now the novelty claim, in place of the "third operational role," and the paper names the object a product of experts.

The second change separates Proposition 7.2 (July 7.3) from the selection bias. Both of you identified the sentence that opened Section 7.2 as invalid; Ezequiel's construction refutes it, and Sara's argument (take the proposal proportional to $w^{-1}P_{true}$) refutes it in general. I deleted the sentence and rebuilt the section as Sections 7.3–7.5: the obstruction in the kernel setting, an example with zero selection bias and an indivisible target, the design rule, and the open problem restated in the form the design rule forces on it.

1. Abstract and significance statement (Sara 1, 2, 5; Ezequiel on originality, Question 9). Accepted. I rewrote both. The significance statement no longer claims uncertainty quantification for the true system; it says the paper quantifies uncertainty under the selected measure and that the target family does not enter Theorems A–C. The contributions now come in two registers: standard results used as such (the SNIS limit theorems, the Chebyshev–Cantelli and Bernstein–Hoeffding bounds, the chi-square characterization of the effective sample size, the Gibbs form, the Chatterjee–Diaconis and Sanz-Alonso scales), and the paper's own (the product identity and its corollary; the finite-grid formulation as forced; the two-tier structure with the vacuity of Tier II in the paper's examples; the conditional transfer paired with the kernel obstruction, with their independence shown by example and the design rule; the quasi-likelihood calibration with the inconsistency of action minimization recorded; two worked examples). That list answers Question 9.
2. Introduction (Sara 6, 10; Ezequiel Finding 8, Question 8). Accepted. I cut the introduction by about a quarter, folded the application classes into the paragraph on the finite grid, and halved the program paragraph. "Quadratic functional" is gone. I cite Donati, Hartmann and Keller (2017) and Adib (2008) in the introduction and in a new Section 10.1. Both reweight an externally generated ensemble by a difference of two actions, equivalently a ratio of two path densities with a common diffusion coefficient; the present construction uses a single action. I draw that distinction, and I confine the two distinctions the July version drew in Section 10.2 to optimization methods, where they belong. That answers Question 8.
3. Section 2, the classical action (Sara 7, 8, 9; Ezequiel on the divergence term, Question 7). Accepted. I define the reference path ξ^o in (4): absolutely continuous, $\xi^o(0) = x_0$, square-integrable derivative, sup-norm tubes, under the Ikeda–Watanabe hypotheses. I narrowed the conclusion of Proposition 2.1 to the drift–diffusion form of the action; a non-Markovian target with a path law may have its own tube asymptotics or rate functional, and the paper says so. I no longer attribute the divergence term to the Itô form of the Girsanov density, which does not display it. It is the Itô–Stratonovich correction one incurs on rewriting the stochastic integral in the Girsanov exponent as a pathwise functional on smooth reference paths, which is the form the tube-limit computation uses; for gradient drift it is $-\frac{1}{2}\int \Delta V \, dt$ from Itô's formula applied to V. That is my answer to Question 7: it is an Itô correction in the sense of the Itô–Stratonovich conversion, and not a term of the Girsanov density.
4. Section 2.2, the degeneracy (Ezequiel Finding 1, Question 6; Sara 15). Accepted. I retitled Proposition 2.3 and narrowed every reading of it to the pointwise weight (6). The text now says that the classical functional on regular reference paths and the Girsanov change of measure on path space are unobstructed, and gives the ratio weight's bounded chi-square on the same grids as the contrast. Question 6: the use of the classical functional as a pointwise importance weight on diffusion samples fails; that use is the continuous-path version of this construction, and it is the only object that fails.

5. Section 3, assumptions and definitions (Sara 14, 20, 21). Accepted. I removed the computable-density condition from the standing assumptions and state it locally in Section 9, where the benchmarks use it; finite chi-square is now (P2). I call formula (7) the left-point discretization of the OM functional, name the kinetic and divergence sums, and say in the sentence after it that (7) is not the negative log Euler density, with a pointer to Proposition 3.5 for the exact relation. Remark 3.7(a) says that different temperatures define different selected measures and that λ is a diffusion-coefficient control; I offer no selection principle for λ beyond the diagnostics of Theorem C.
6. Section 3.4, the identity (Ezequiel Finding 2, Questions 1 and 2; Sara 12). Accepted; this is the change described above. Proposition 3.5 gives the identity for general surrogates, e^{−λS} = (2πδt)^{λdM/2} |det σ_v|^{λM} p_v^λ e^{−λS^div}, and its constant-divergence form. Corollary 3.6 gives the Gaussian case with the exact and leading-order relations, the Markov property, and σ_eff < σ_Π for all λ > 0. Question 1: the construction multiplies because it reweights by a single action, which is a density and not a ratio of densities; the paper now states that this is the point of departure and the source of the construction's behavior, and Section 11 says when a product is what one wants (when proposal and surrogate are each partially informative and neither is a reference measure for the other). I do not claim a product is preferable in general. Question 2: the derivation is the proof of Corollary 3.6(ii); the consequence for the quadratic variation is the last sentence of the corollary. I accept Sara's item 12 with a correction to my own first draft of the revision: for a Gaussian Markov proposal the selected measure is Markov; for a Gaussian proposal with longer-range dependence it inherits that dependence, and Example 7.4 is a Gaussian, indivisible instance. I weakened the claim that the selected measure "cannot come from any generator-based model" to "is not established as the law of a generator-based model reproducing the target family."
7. Section 3.5, well-posedness and the estimator (Sara 3, 11, 13, 16). Accepted. Lemma 3.9 (July 3.6) carries Sara's sharper bound $\chi^2 \le W_{max}/Z_v - 1$; in the example it reads 3.5×10^{7} against 1.2×10^{15} before and 1372 exact, and the text says the bound certifies finiteness and nothing more. I replaced the statement that w dΠ "is not the unnormalized posterior of any generator-based Bayesian model" as in item 6. Remark 3.11 (July 3.8) says the generalized-Bayes resemblance is algebraic only: the candidate path is the variable distributed under Q, no data condition it, and observed data enter through calibration, which is a separate operation. Remark 3.12 (July 3.9) says the causal factorization of the weight licenses a sequential implementation only when the proposal itself admits compatible sequential sampling.
8. Section 4 (Sara 17, 18; Ezequiel on Theorem 4.1(iii), Question 10). Accepted. I removed "minimal moment conditions"; the text after Theorem 4.1 says the hypotheses are standard sufficient conditions and that integrability of the denominator with finite variance of (F − F*)w suffices. Remark 4.2 says that I supply no primitive sufficient condition for the uniform integrability in part (iii), that in the heavy-tailed regime it is not a formality, and that the asymptotic mean-squared-error statement is unavailable in practice unless one establishes such a condition separately. That is my answer to Question 10: none is available, and the paper says so.
9. Section 5 (Sara 19; Ezequiel on Remark 5.4). Accepted. A sentence after Theorem 5.1 says the bounds can exceed the trivial ones when N is small relative to χ^2 and are informative once $N \gg \chi^2$. Remark 5.4 keeps the observation that Tier II is vacuous in the first example, gives the exponent (4.2×10^{-16} N), and now attributes it to the product structure: Z_v is the overlap of two path densities and is exponentially small in M.
10. Section 6 (Sara 27, 28; Ezequiel on the thresholds). Accepted. I label the 10^{2} and 10^{-2} thresholds of Remark 6.2 as heuristics that follow from nothing in the paper, ask that the Pareto shape estimate $\hat{k}$ accompany them, and say that smoothing the weights changes the estimator to which Theorems A–C apply. I qualify "variance-inflation factor" as an observable-independent upper-bound factor.
11. Sections 7.1–7.2, the two settings and Theorem D (Sara 29, 37). Accepted. Section 7.1 defines the kernel setting (K), in which only the two-time family is given and general path observables are not identified, and the process setting (L), in which the inducing process and its grid law are specified. Theorems A–C belong to neither; Theorem 7.1 lives in (L); Proposition 7.2 in (K); Table 3 records the setting of each row. The tolerance in (D1) is "established or estimated by means external to this paper," and the paragraph after the theorem says that in an application with unknown path law one cannot test (D1) from proposal samples, and the estimator's output is F*(v).
12. Sections 7.3–7.4, the obstruction and the selection bias (Ezequiel Finding 3, Question 3; Sara 4, 22, 30, 31). Accepted. I deleted the sentence "the conditional structure of Theorem D is forced, not chosen" and state its withdrawal. Section 7.4 explains why the inference fails: Q is a product of the proposal law and the tempered surrogate density, the proposal is free, and separation of the surrogate's kernels from the target's is independent of separation of the selected measure from the target's path law. Example 7.4 is Ezequiel's construction: M = 8, k_v = 2, λ = 1/2, the proposal precision with the two ε = 2.5 cross terms (least eigenvalue 0.339), the target with precision K_Π + λK_v (least eigenvalue 1.270), Q = P_true exactly, and no triple composing: at (3,5,7) the direct coefficient is −0.008588 against a composed 0.305127, an absolute gap of 0.314, and the smallest gap over the 56 triples is 0.011. I added one sentence that neither review states: the construction sets K_Π = K_target − λK_v, so the proposal attains zero bias by presupposing the target law. That is the scope of the obstruction and my answer to Question 3: improving the surrogate's kernels cannot close the truth link, and the proposal can close it only by presupposing the target law. Sara's item 30 is Remark 7.3: μ invariant for the surrogate semigroup suffices for the contraction, and for the OU surrogate this is its stationary Gaussian law, which the target's reference measure must then be. Item 31 is in Section 8.3: the obstruction bounds the closure uniformly in operator norm at the worst pair and says nothing about other metrics, particular observables, restricted pairs, or the small-deficit regime.
13. Section 7.5, the design rule and the open bridge (Ezequiel Finding 4, Questions 4 and 5; Sara 23). I adopt the first of Ezequiel's two remedies and not the second (item 21). Proposition 7.5 states that for Gaussian pairs,

as $M \to \infty$ at fixed $(\lambda, \sigma_\Pi, \sigma_\nu)$, the total-variation distance to a target of diffusion coefficient σ^2 tends to one unless $1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2 = 1/\sigma^2$; with $\sigma_\nu = \sigma$ this requires $\sigma_\Pi^2 = \sigma^2/(1-\lambda)$, impossible at $\lambda \geq 1$, the temperature of the first example. It is a necessary condition, proved for Gaussian pairs; I expect the same mechanism for general constant-diffusion pairs and say in the text that I have not proved the extension. I reproduce Ezequiel's Bhattacharyya table ($d_{TV} \geq 0.527, 0.774, 0.948, 0.9973, 0.99999$ at $M = 25$ to 400). Remark 7.6 restates the open problem: no invariant of the target can control a quantity that depends on the proposal–surrogate–temperature triple, so the question is a design problem at fixed M over the class of triples whose selected quadratic variation matches the target's, with proposal and surrogate errors measured separately. The proposition answers Question 4: (D1) is attainable under refinement only on that class, and the triple controls η, not the target alone. Question 5: no. Holding χ^2 fixed at 10 or at 1372.4 and solving for λ_M gives displacements of the estimand from its proposal value of $0.1903 \to 0.0569$ and $0.2800 \to 0.0943$ over $M = 50$ to 800 (Section 11, your table reproduced), so the schedule drives the construction to the proposal. The selected measure itself converges: at fixed λ the estimand goes from -0.839027 at $M = 25$ to -0.844743 at $M = 6400$, with limit -0.84476 by Richardson extrapolation, the terminal readout of the OU law with $\sigma_{eff}^2 = 1/2$ and $k_{eff} = \sqrt{(5/2)}$ tilted at the endpoint. The estimator has no limit because the limit law is singular to the proposal's. The paper says both.

14. Sections 8.1–8.2, the (P2) certificate. Added. Lemma 8.1: if $\nabla \cdot b \geq -c_2|x|^2 - c_0$ and the proposal is Gaussian with covariance Σ_Π, then (P2) holds whenever $\lambda c_2 \delta t \|\Sigma_\Pi\| < 1/2$. The certificate discards the kinetic sum and is crude; for the double well it holds at $\lambda = 3/4$ (0.452) and fails at $\lambda = 1$ (0.602), so the second example uses $\lambda = 3/4$.
15. Section 8.3, Mori–Zwanzig (Sara 26, 31). Accepted. I wrote the closure out. The memory kernel collapses to $\gamma = \int \kappa$; I model the fluctuating force as white noise with intensity $\sigma\sigma^\intercal = \int E[f(t)f(0)^\intercal]dt$, which the fluctuation–dissipation relation evaluates to $\gamma C + C\gamma^\intercal$ in the Mori case; the surrogate pair is $b = \Phi(u) - \gamma u$ with that $\sigma\sigma^\intercal$. The intensity is positive semidefinite by Bochner's theorem; non-degeneracy is an assumption on the projection, and I state it as one. The July text said the fluctuating force is dropped; that leaves $\sigma = 0$, which is not a surrogate in the sense of Definition 3.3, and the text now says the July description was wrong.
16. Section 8.4, calibration (Ezequiel Finding 5, Question 11; Sara 24, 25, 44). Accepted, with one addition. I withdraw minimizing the empirical action as the calibration procedure. It is inconsistent even when the truth is in the family: with the observed law equal to the OU proposal ($k = 1$), $E_\Pi[S_k] = (k-1)^2(\delta t/2)\Sigma_m \mathrm{Var}(\xi_m) + M/2 - kT/2$, minimized at $k^* = 1 + [2\delta t\, \Sigma_m \mathrm{Var}(\xi_m)]^{-1} = 2.76$. The divergence term $-kT/2$ shifts the minimizer; the kinetic sum alone has its minimum at $k = 1.000$. It is also ill-posed in σ_ν, as Sara showed. Calibration is now Euler quasi-likelihood M-estimation (Florens-Zmirou, Kessler, and Uchida–Yoshida for the misspecified case): the objective has an interior minimizer in σ, is consistent when the observed law is the surrogate's Euler chain, and otherwise converges to the KL projection of the observed grid law onto the surrogate family. I accept Sara's item 24 as stated: a KL projection needs a true probability measure, which setting (L) supplies, and not a generator. I distinguish three parameters and reserve "optimal" for the second: ν_{QML} (calibrated; $k = 1.000$ in the example), $\nu^\dagger = \mathrm{argmin}\, \sigma^2(\nu)$ for a fixed observable ($k = 12.39$, $\sigma^2 = 4.999$), and the withdrawn ν^* ($k = 2.76$, $\sigma^2 = 20.3$). For a class of observables one must specify the objective (Sara 44), and the text says sup or average. The closed form above answers Question 11. Selection still uses the OM action at the calibrated ν; I separate the two operations deliberately (Sara 11).
17. Section 9, the examples (Ezequiel Findings 6 and 7, Questions 13 and 14; Sara 33, 40, 41, 42). Accepted. I reran the convergence study at 1,000 replications with a fixed seed, with a bootstrap standard error on $\sqrt{N}\cdot$std and the range of that statistic over the twenty disjoint batches of fifty. At $N = 10^4$ the range over the twenty batches is 2.66 to 4.75, wider than your 3.30 to 4.57 over ten, and at $N = 10^5$ the pooled value is 5.29 ± 0.17 against the asymptote 5.56. The new table supports the July caption's claim; the July table did not. The plug-in $\hat{\chi}^2$ on this draw is 1059, against 623 on the July draw and 1372.4 exact; the text reports the variability. The table and the data statement answer Question 14: I deposit the script, seeds and output with the preprint. The second example is the double well $b = x - x^3$ against the same proposal at $\lambda = 3/4$. The benchmark is deterministic: the unnormalized selected density is a product of one-step kernels, and I obtain Z, F^*, χ^2, D_{KL} and σ^2 by propagating densities through fifty one-step integral operators on a mesh, stable to six figures under mesh refinement. With the divergence term, $F^* = -0.547760$, $\chi^2 = 162.1$, $\sigma^2 = 44.31$; without it, $F^* = -0.637163$, $\chi^2 = 169.3$, $\sigma^2 = 26.60$. The term moves the estimand by 0.089 and reverses the sign of the selection displacement from the proposal's value -0.561929. That answers Question 13. The double well is the only example in the paper in which the divergence term acts, and that term is what separates the OM action from the Euler exponent. Table 2 is the 1,000-replication study for the double well. Sara's items 33, 40 and 41 are in the captions and in Section 9.4: the example benchmarks the estimator and does not validate the truth link; the rate 0.144 is specific to the parameters; the figure illustrates and does not verify.
18. Section 9.6, grid dependence (Ezequiel's [R] item, Questions 12 and 15). Accepted. A new subsection says that M, λ, the proposal and the discretization convention are part of the definition of the estimand, that I establish no invariance under the left-point/midpoint convention, coordinate change or nested grids, and that empirical use must report sensitivity to all four. Question 12: in the Gaussian example the M-dependence is an $O(\delta t)$ discretization error toward the explicit limit of item 13, but the family $\{Q_M\}$ is not projectively consistent and I have no telescoping identity across grids. Question 15: for state-dependent σ the classical action acquires convention-dependent terms; the restriction to constant σ_ν is what keeps (7) free of them, and I leave the state-dependent case out of scope.
19. Section 10, the comparison (Sara 34, 35, 36, 38; Ezequiel Finding 8). Accepted. Section 10.1 is new and positions the construction against Girsanov path reweighting as in item 2. In Section 10.2 I withdraw the

claim that misspecified-model theory needs a generator-based P0; the calibration of Section 8.4 is an instance of that theory and I present it as one, and the distinction that survives is that the selection step is not a fit. In Section 10.3 I withdraw the claim that path-space relative entropy presupposes a generator-derived density; the distinction that survives is that the variational constructions optimize the surrogate toward the target and need the target's law to do so. In Section 10.4 I withdraw the PAC-Bayes transfer: $\hat{\nu}_{QML}$ is a point estimator, and I formulate no prior on Θ, posterior over ν or empirical-risk setting. I rebuilt Table 3 with rows for setting, weight form, quadratic variation, chi-square under refinement, fitting target and truth link; its caption says which July distinctions I dropped and why.

20. Section 11 (Sara 39, 43, 45, 46; Ezequiel's [R] items on the temperature schedule and the estimand, Question 16). Accepted, with one problem closed. I close the continuum problem as in item 13 and state it as a conclusion: the selected measure has a limit, its importance-sampling representation from the proposal does not, and the only renormalization I know that restores absolute continuity is subtraction of the proposal's kinetic term, which is Girsanov reweighting. The multilevel problem is conditional: the limit exists and the level bias is $O(\delta t)$, but the level-wise chi-square grows as $e^{0.144M}$ and the selected measures are not projectively consistent, so the coupling is what is missing. I call the risk expression the asymptotic trade-off and not a risk formula; the observable-class optimum requires a stated objective; I state adaptive-proposal consistency as what would need to be proved. I can answer Question 16 least well. When (D1) cannot be tested, $F^*(\nu)$ is the expectation under a product of experts of the proposal law and the tempered surrogate law, consumed as a point readout where such a consensus is what the application wants, at a coarse grid, with the diagnostics of Theorem C and the sensitivities of Section 9.6 reported alongside. Its decision-theoretic value is application-specific and the paper does not assert one.
21. Not adopted. Ezequiel's alternative remedy in Finding 4, modifying the construction so that the selected measure preserves the proposal's quadratic variation, is subtraction of a reference kinetic term; it converts the weight into a Girsanov ratio and the paper into a description of an established method. I kept the construction and stated what it is. Sara's alternative for the Mori–Zwanzig section, presenting it as a possible source of surrogates rather than a characterized family, was not needed once I wrote the closure out. On the assessment that the component theorems are standard and the originality is moderate, I have no counterargument; the contributions list says so, and the paper's own material is the identity, the finite-grid character as forced, the two-tier structure, the independence of the obstruction from the bias, the design rule, the calibration result and the two examples.

Bibliography: I added Donati–Hartmann–Keller 2017, Adib 2008, Hinton 2002, Kessler 1997, Florens-Zmirou 1989, Zwanzig 2001, Tsybakov 2009 and Brockwell–Davis 1991, and removed Alquier–Ridgway–Chopin with the PAC-Bayes paragraph. The July entry for "Stochastic optimal control matching" attributed the paper to Holdijk et al.; it is by Domingo-Enrich, Han, Amos, Bruna and Chen (NeurIPS 2024), and I corrected it. Chao–Duan now cites the Nonlinearity version and Huang–Zhou–Duan the SIAM DOI. I updated the companion citations: the foundations paper under its new title, "The Physics of the Instant," and the projection paper with its SSRN DOI. Every result cited from the Koopman paper retains its number in revision 4.

Let me know your thoughts.

Ben

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